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Objective

The goal of this project is to build a reliable charging system that converts AC power into a steady 14.4V DC output for an E-bike battery. To test the design, a programmable power supply was used to mimic a wheel-mounted generator. The system uses a full-wave rectifier, an inverting buck-boost converter, and an Arduino-based PI control loop to keep the output stable and safe.

Overview

- Demonstrates a current PI control loop to drive converter for desired output current.
- Inverting Buck-Boost converter is used to regulate output voltage and current depending on state of charge of battery.
- Current sensor and voltage divider are used to send voltage and current information to an Arduino Uno R4.

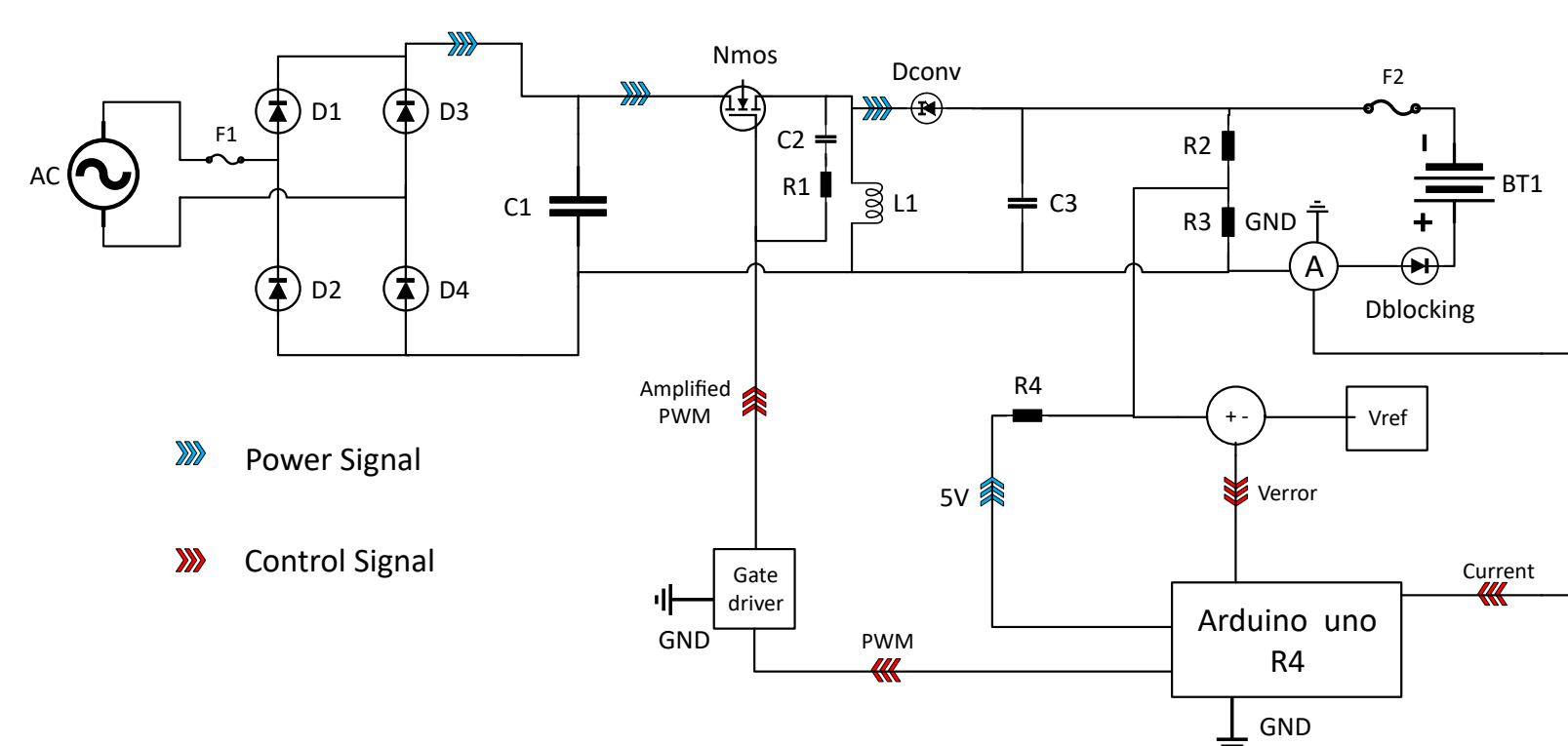


Fig 1: E-Bike Battery Charging System circuit schematic.

- The goal is to control a varying AC input to a regulated DC output while obtaining the ability through closed loop control to charge a battery.
- Minimum converter capacitor value is estimated by

$$C_{min} = \frac{I_{out} \times D}{F_s \times V_{pp}}$$

- Minimum converter inductor value is estimated by

$$L_{min} = \frac{(V_{in} \times D)}{2 \times I_{out} \times F_s}$$

System Implementation

- The Arduino Uno R4 executes a multi-stage software routine to manage a proportional-integral feedback loop at 30kHz-100kHz switching frequency.
- Signal Conditioning: Processes raw analog data (A0 for current, A1 for voltage) and utilizes a Digital Low-Pass filter. This dampens reading fluctuations over time preventing erratic noise or data fluctuations from causing instability in the control loop.
- CC/CV State Machine: Manages transition between Constant Current (CC) and Constant Voltage (CV) charging phases based on the batteries state of charge.
- Integrator Anti-Windup: Restricts error accumulation by clamping integral values between 0.2 and 0.55. This prevents runaway calculations and keeps the physical hardware within safety limits.
- Firmware Safety and Protection loops: Continuously checks for output voltage collapse and analog input pin over-voltage.

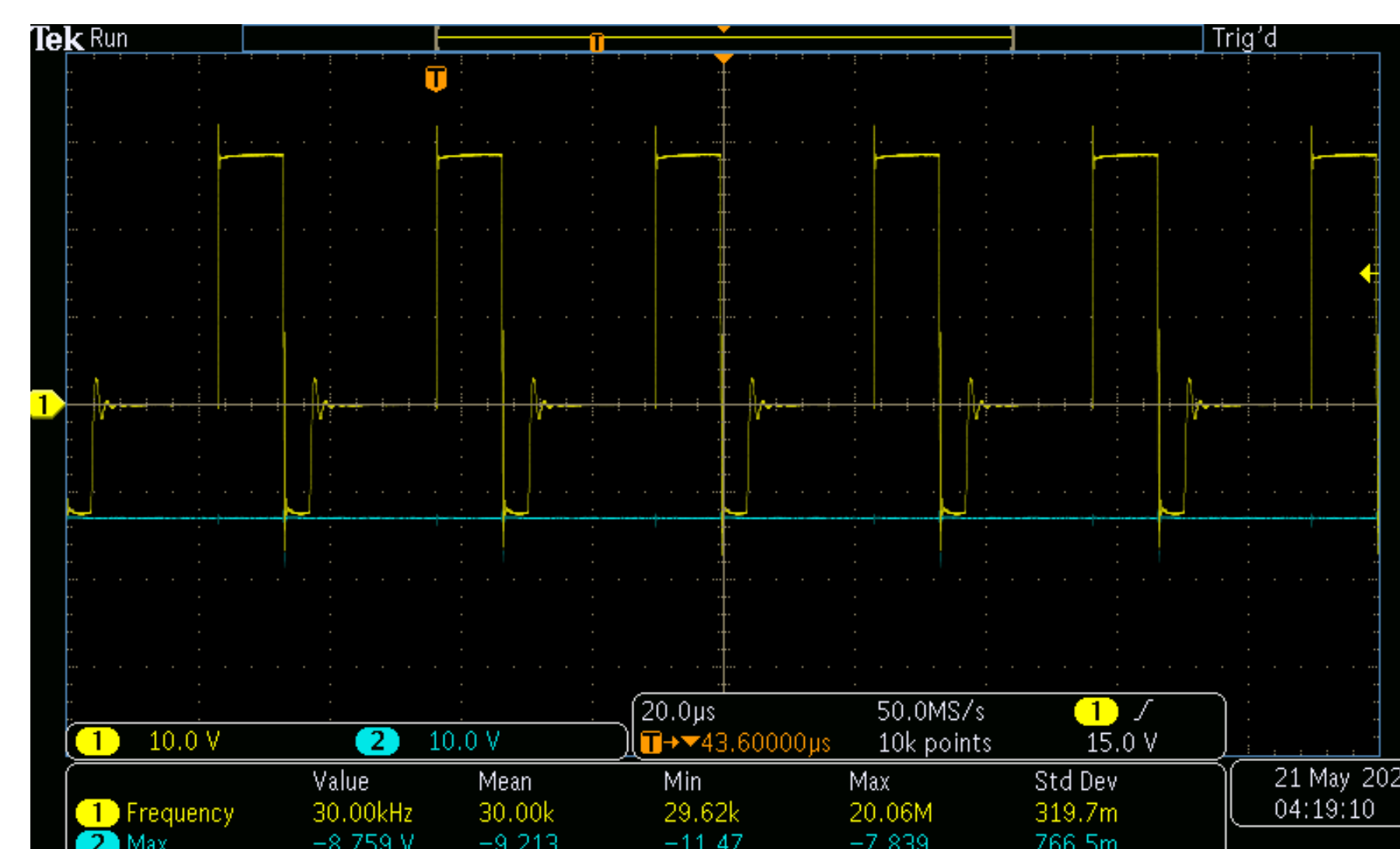


Fig 2: Oscilloscope shot showing the PWM output of gate driver (yellow) and a corresponding voltage output (blue) dependent on duty cycle and voltage input

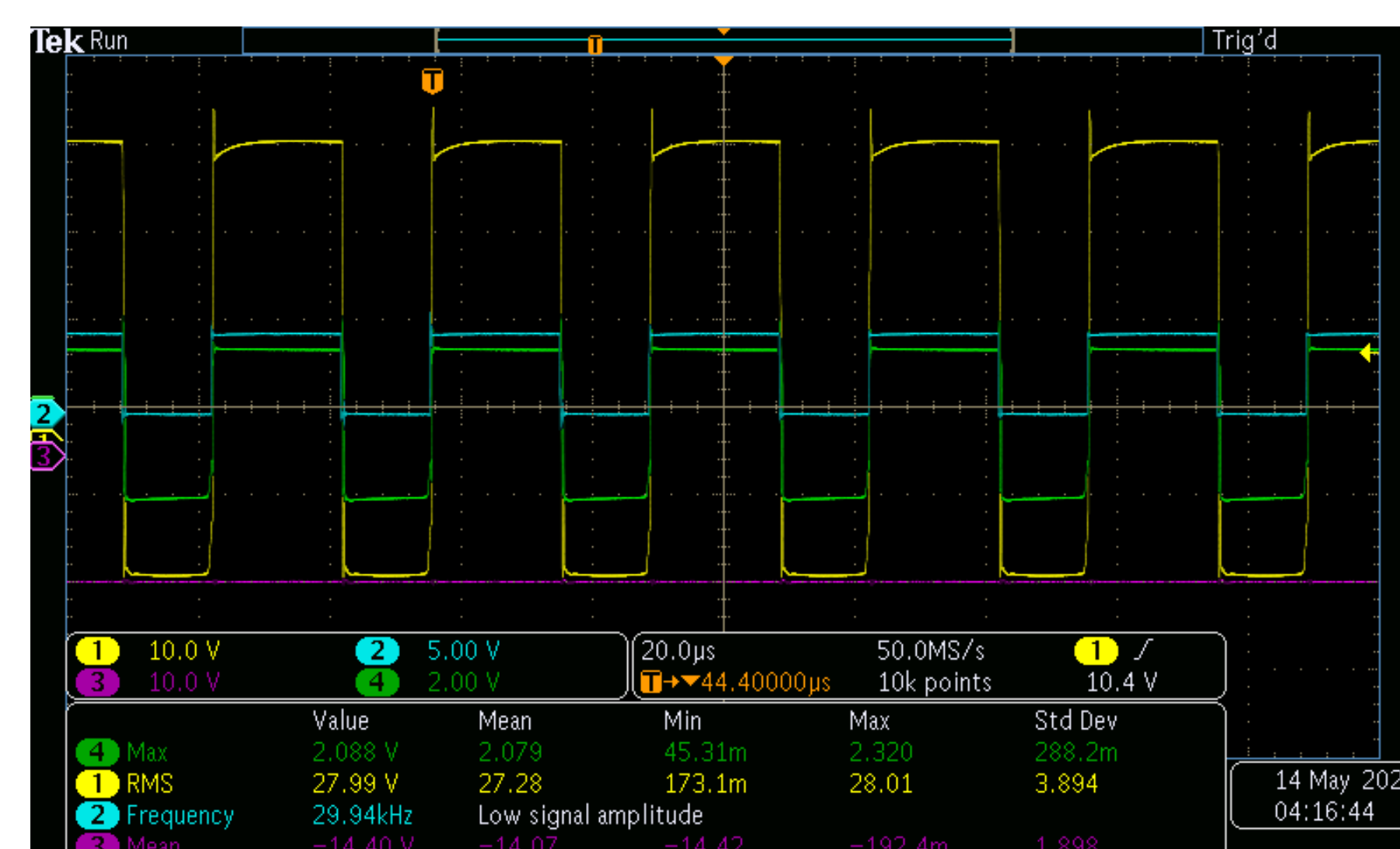


Fig 3: Oscilloscope shot showing the PWM output of gate driver (yellow), PWM input to driver from Arduino (blue), source node of MOSFET (green), and voltage output of converter (purple).

- Open-Loop hardware verification
- Verifies gate driver can output a stable PWM at the correct frequency.
- Verifies the PWM can drive the MOSFET.
- Verifies the converter successfully converts voltage in to a new voltage out.

- Closed-Loop control verification

- Verifies the control-loop stability at a target voltage out while input voltage varies.
- Verifies the absence of noise/oscillations causing control-loop instability.
- Verifies output voltage recovery when input voltage varies above or below input thresholds and returns within those thresholds.

Simulation

- The Simulink simulation provides a realistic simulation environment in which we can test power designs and control loop stability.

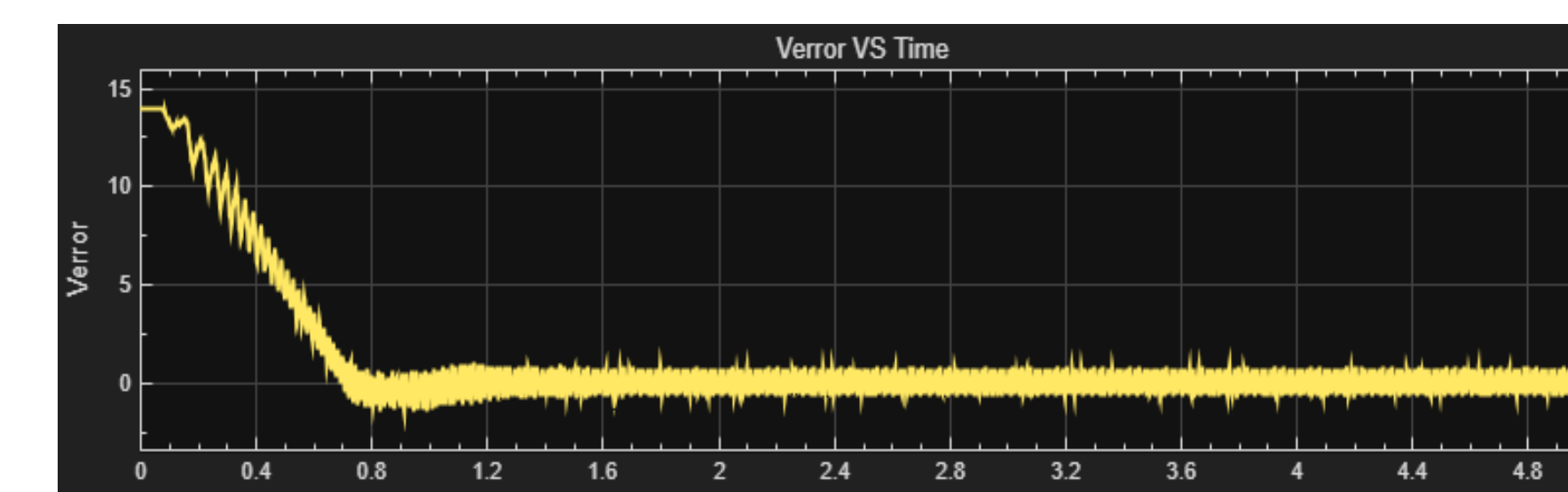


Fig 4: Voltage error versus time plot in simulation

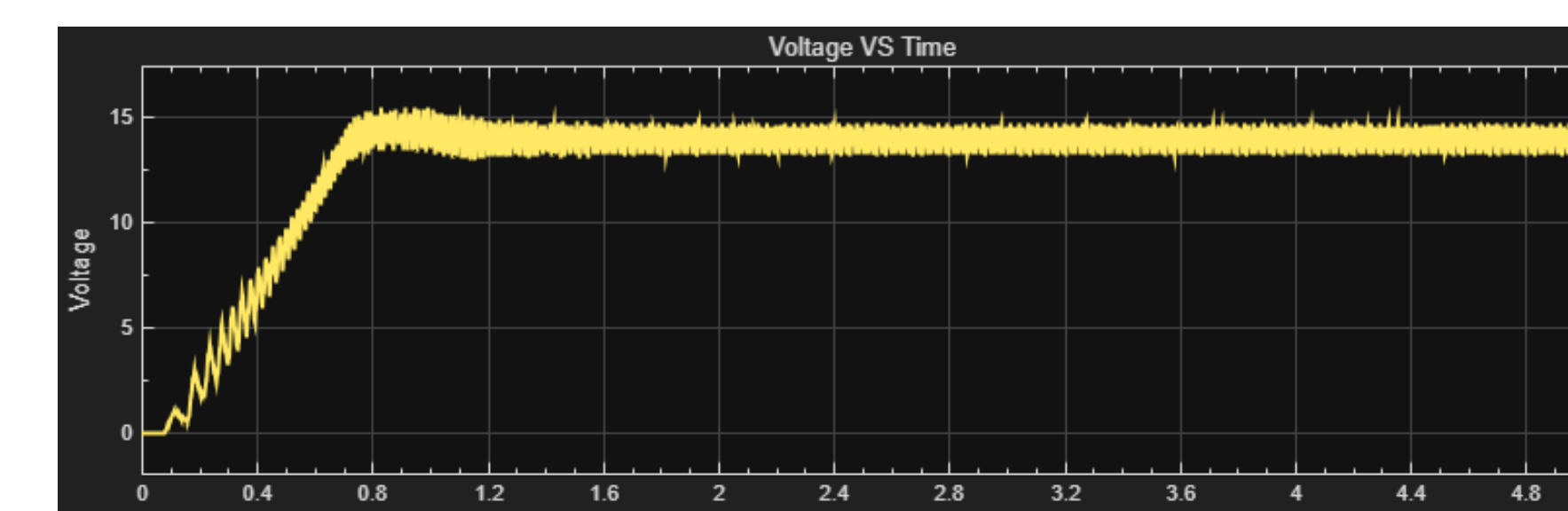


Fig 5: Voltage output versus time plot in simulation.

- Voltage output and voltage error graphs show stability and control through proportional integral control loop.
- Voltage error starts at initial voltage target and achieves a settling time of 0.8 seconds.
- Voltage out starts at 0V and climbs to target voltage of 14.4V. PI control achieves a settling time of 0.8 seconds.
- Simulation verifies the goal of minimal overshoot to ensure battery safety and prevent overvoltage of battery.
- Voltage ripple of 1.6V on average which indicates an increase of converter output capacitor size is needed.

Acknowledgements

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